9. Proactive Cybersecurity

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Emerging Security Issues: outline

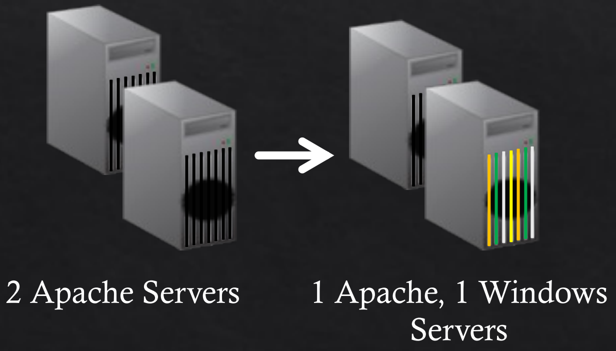
- Emerging Security Issues
  - Emerging networks
    - cloud, SDN, IoT
  - Vehicle security
  - Homomorphic encryption
  - Security and Humans

Moving Target Defense

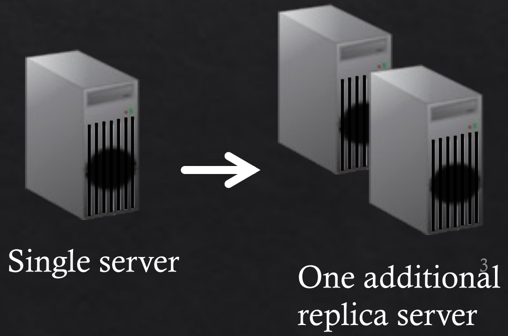
- Simply MTD, is a defense mechanism that continuously changes the attack surface to thwart cyberattacks
- Different types of changes can be made in the system, and they can be categorised into three

Shuffle

Diversity



Redundancy



MTD

- MTD can also be implemented in different layers of the system
  - For example:
    - Application layer – code regeneration/diversification
    - Transport layer – routing node redundancy
    - Internet layer – IP shuffling and virtualisation

MTD

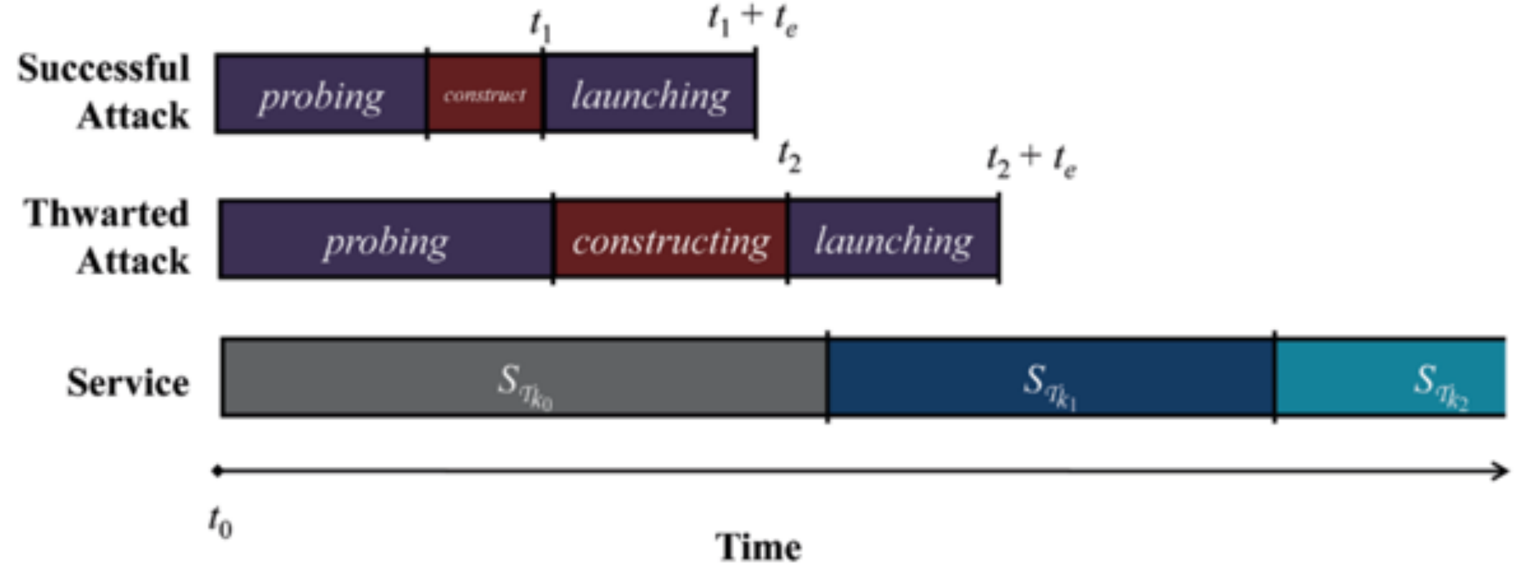


Fig. 2.1 Attack Lifetime

Evans, David, Anh Nguyen-Tuong, and John Knight. "Effectiveness of moving target defenses." *Moving Target Defense*. Springer New York, 2011. 29-48.

MTD Shuffle

- The MTD shuffle technique **changes** the configuration of the system such that connections/dependencies are altered but the operations are maintained
  - E.g., changing the routing table in the SDN
- Existing vulnerabilities are **not removed**, but it redirects the attack sequence to be taken
  - Hence, delaying the reconnaissance process, as well as breaking an ongoing attack

MTD Shuffle

- Example: Openflow Random Host Mutation
  - Typically, a host has one (physically) IP address
  - In HF-RHM scheme, each host has a physical IP address and a virtual IP address
  - Only the SDN controller knows the physical IP address of each host
  - Virtual IP (vIP) addresses are used for establishing communication channels
  - vIP addresses are shuffled every x seconds, OF switches are updated to redirect flows and update the existing communications
  - Attackers probing for IP addresses loses the identified victim host as vIP shuffles

Jafarian, Jafar Haadi, Ehab Al-Shaer, and Qi Duan. "Openflow random host mutation: transparent moving target defense using software defined networking." *Proceedings of the first workshop on Hot topics in software defined networks*. ACM, 2012.

MTD Shuffle

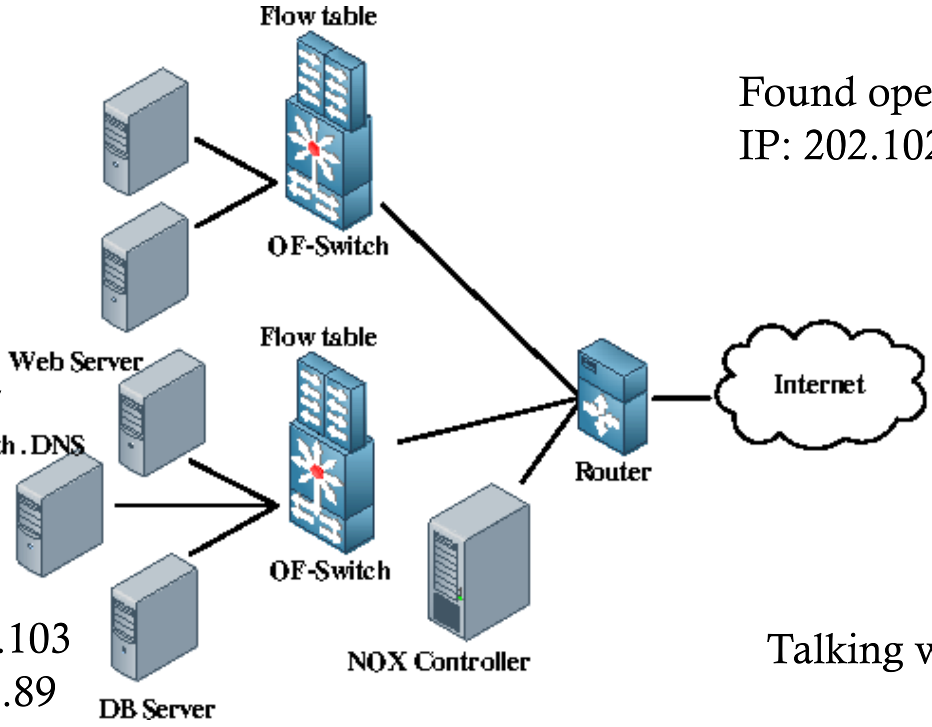
1 Attacker found vulnerabilities in the web server by scanning

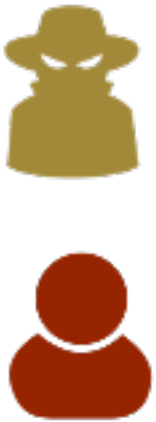
Found open port at IP: 202.102.75.22

rIP: 192.168.0.33
vIP: 202.102.75.22

rIP: 192.168.0.32
vIP: 202.100.64.84

rIP: 192.168.0.103
vIP: 202.116.3.89





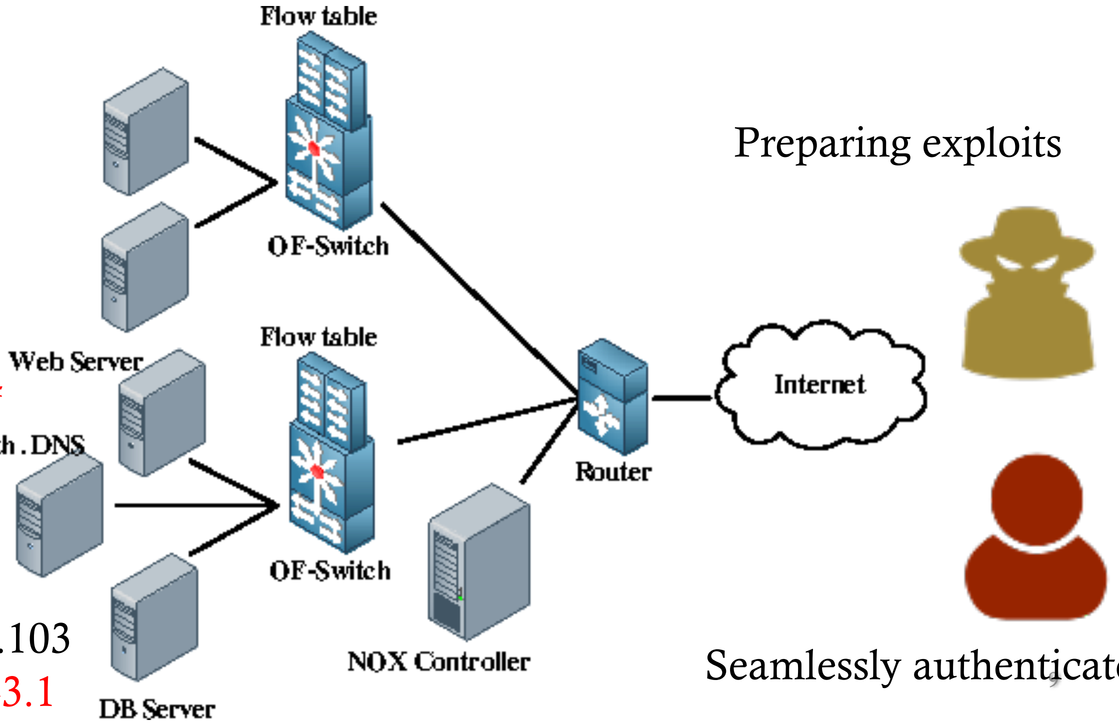
Talking with web server

MTD Shuffle

2

SDN controller shuffles vIP

Preparing exploits



rIP: 192.168.0.33
vIP: 202.122.78.14

rIP: 192.168.0.32
vIP: 202.167.69.41

rIP: 192.168.0.103
vIP: 202.198.43.1

Seamlessly authenticate

MTD Shuffle

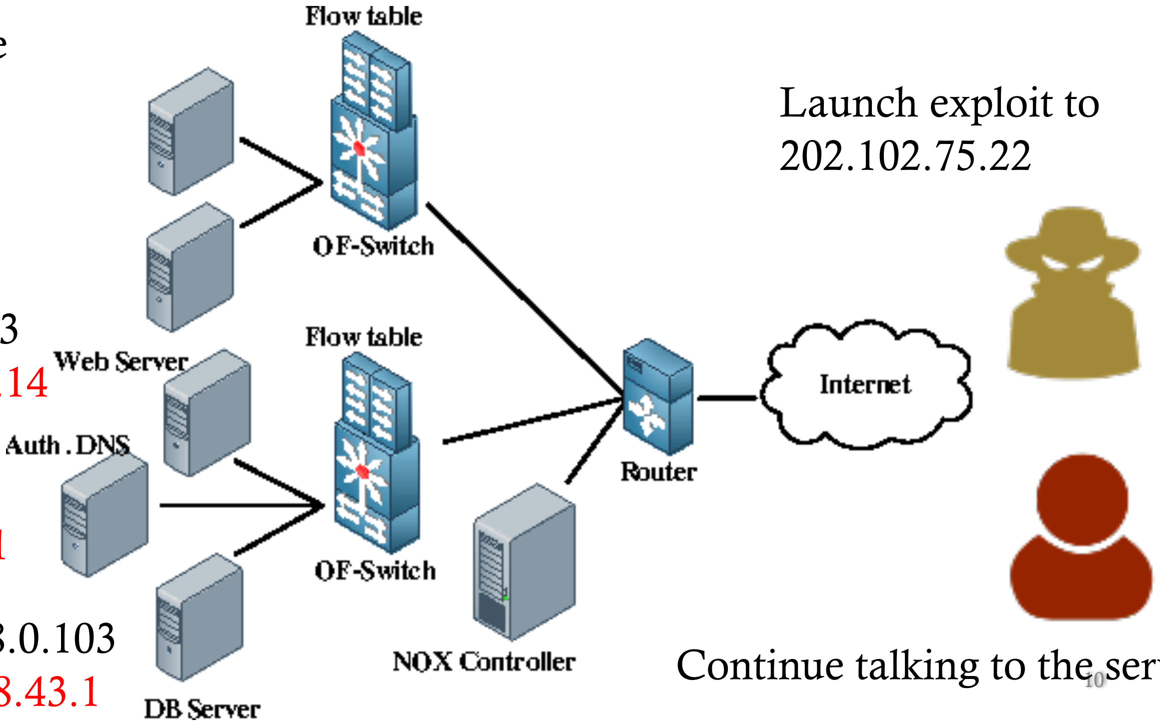
3

Attacker cannot reach the target

rIP: 192.168.0.33
vIP: 202.122.78.14

rIP: 192.168.0.32
vIP: 202.167.69.41

rIP: 192.168.0.103
vIP: 202.198.43.1



Launch exploit to 202.102.75.22

Continue talking to the server

MTD Shuffle

- What are some issues to consider using shuffle?
  - Shuffled system state may not be secure
  - Shuffle may violate security requirements
  - Shuffle may not be feasible
    - E.g., cannot satisfy performance requirements
  - Shuffle may not be applicable
    - E.g., homogeneous network/system
  - How frequently should we shuffle?
    - Proactively, reactively, etc.

MTD Diversity

- Diversity technique provides replacement of an existing component/service, with **different implementation/configuration** that provides the **same functionalities**
- Existing vulnerabilities may be **replaced** with a new set of vulnerabilities
  - E.g., replaced the Windows-based computer to Mac

MTD Diversity

- Example: Compiler generated software diversity
  - A version of software has a specific signature/binary
  - There are number of instructions to carry out the software task
  - These instructions can be remapped with other (single or combinations) instructions to do the same operations
    - Simply, 3+5 is the same as 14-6

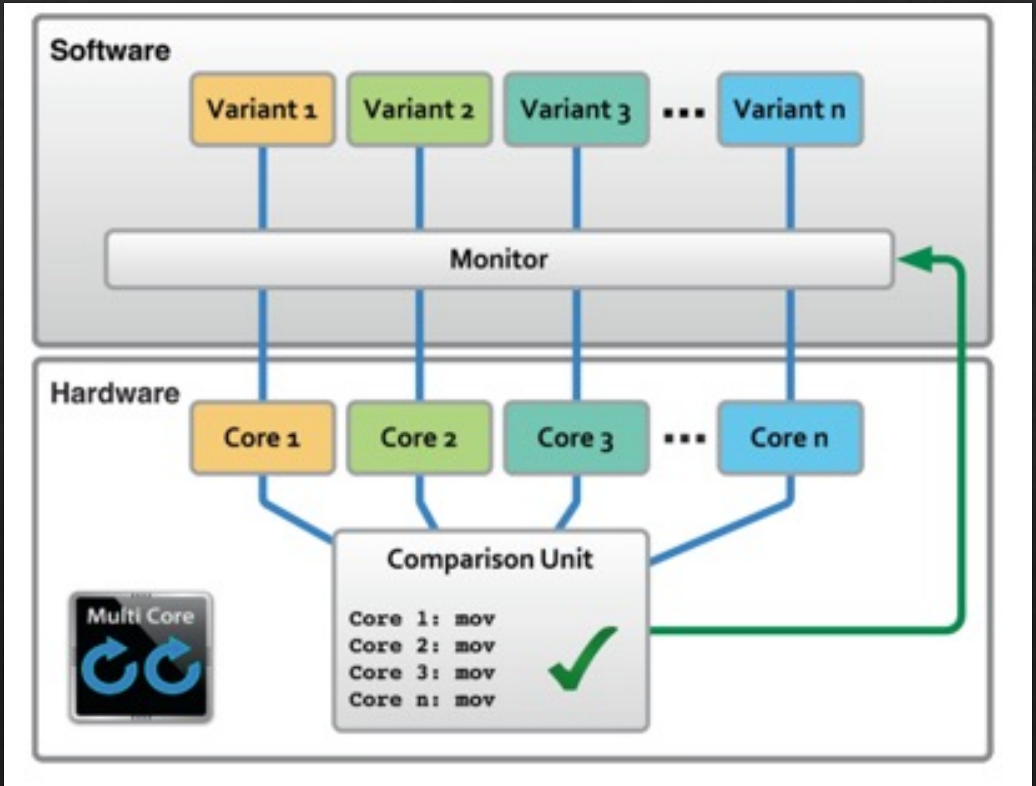
```
movl   %edx, %eax
xchgl  %edx, %eax
```

can be replaced with

```
leal   (%edx), %eax
xchgl  %edx, %eax
```


Jackson, Todd, et al. "Compiler-generated software diversity." *Moving Target Defense*. Springer, New York, NY, 2011. 77-98.

MTD Diversity



Variant monitor and comparison units are added to ensure the same operations by the variant software

Jackson, Todd, et al. "Compiler-generated software diversity." *Moving Target Defense*. Springer, New York, NY, 2011. 77-98.

MTD Diversity

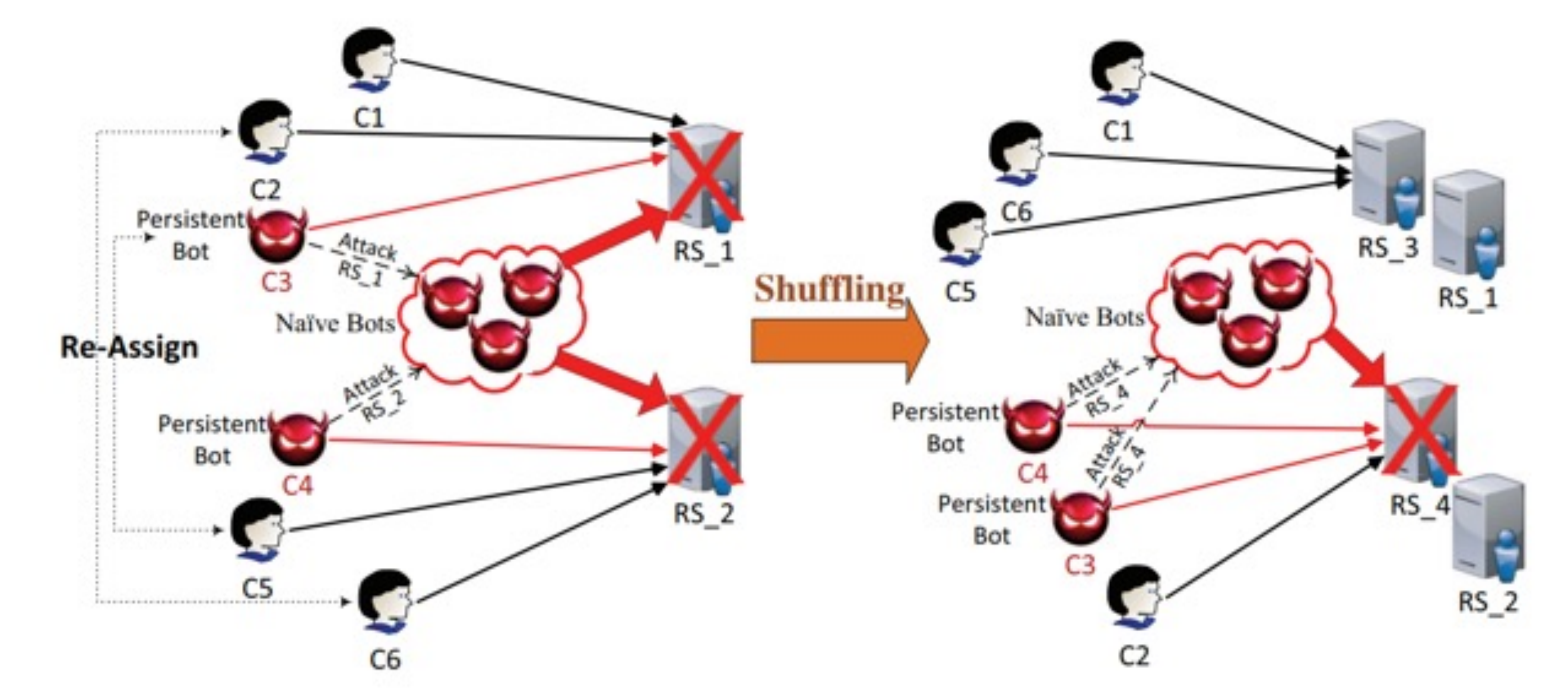
- What are some issues to consider using diversity?
  - New variant(s) may be more vulnerable
  - Difficult to generate variants
  - Cost to implement diversity technique
    - E.g., OS diversity in servers
  - Downtime is more significant than shuffle
    - E.g., rebooting vs SDN flow switching

MTD Redundancy

- Redundancy aims to ensure the availability of the system
- Replicas provide resources to handle high usage of the system
- It can be used in conjunction with other MTD techniques to provide defense against attackers
- For example DDoS attacks

Jia, Quan, et al. "Catch me if you can: a cloud-enabled DDoS defense." *Dependable Systems and Networks (DSN), 2014 44th Annual IEEE/IFIP International Conference on*. IEEE, 2014.

MTD Redundancy



Jia, Quan, et al. "Catch me if you can: a cloud-enabled DDoS defense." *Dependable Systems and Networks (DSN), 2014 44th Annual IEEE/IFIP International Conference on*. IEEE, 2014.

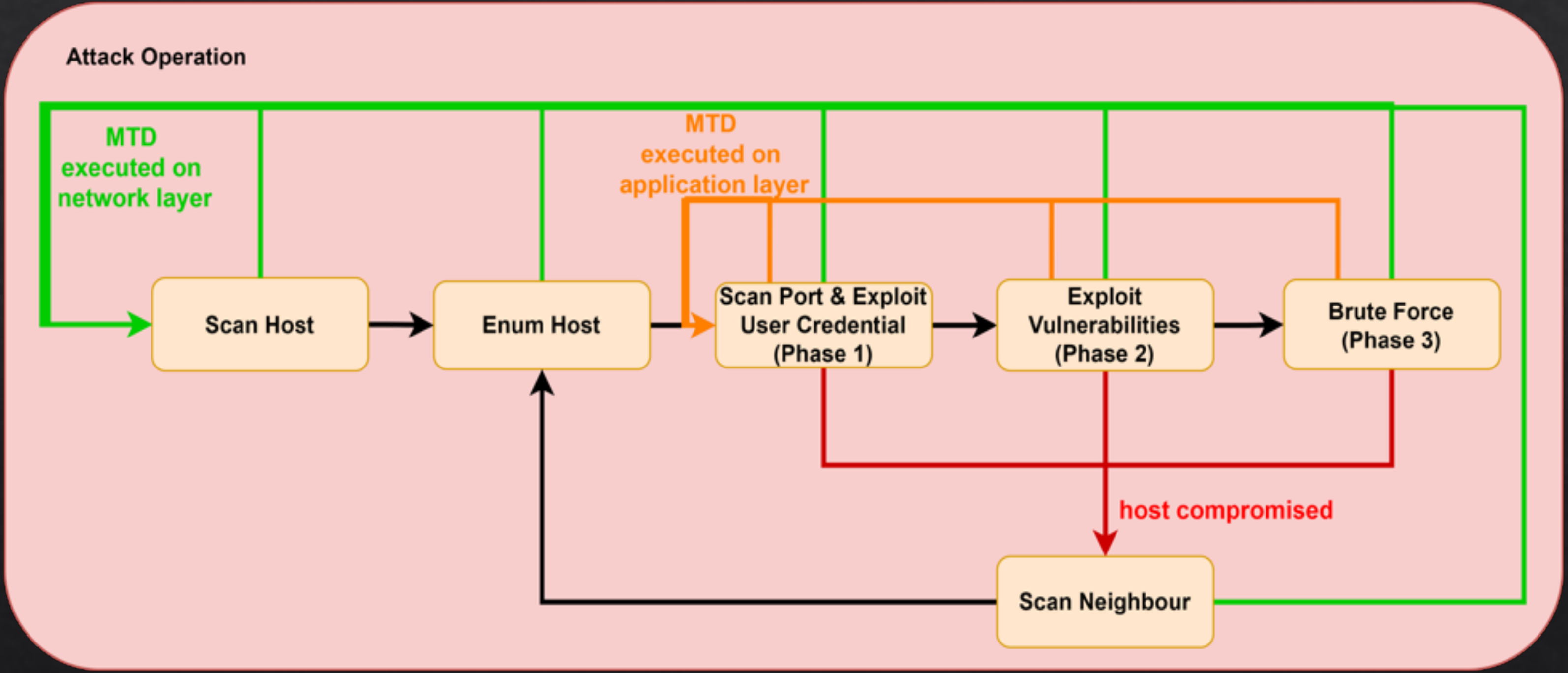
MTD Redundancy

- By adding redundant servers and shuffling identified bots to the same server, we can ensure the **availability** of the cloud resources to legitimate users
- Problems with redundancy techniques include
  - Resource constraints
  - Ineffective against attacks violating confidentiality and integrity (but could be used for tolerance)
  - Degrading performance
    - Increasing redundancy does not proportionally increase availability

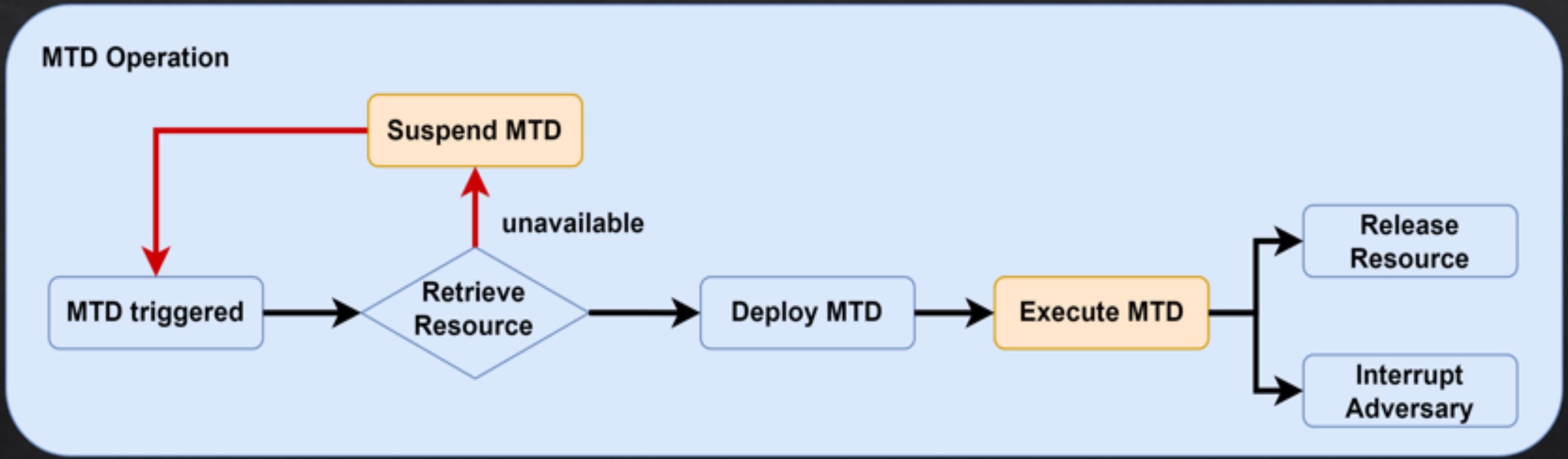
What next in MTD?

- Different MTD techniques provide **different** approaches to ensure the **security objectives** of the system
- Many MTD research has been conducted in the last decade
- It is difficult to evaluate how effective MTD is in your own system
- It is uncertain how to optimise MTD deployment scenario
  - Is one MTD technique enough?
  - Are two MTD techniques better than one?
  - Can I randomly deploy MTD techniques?

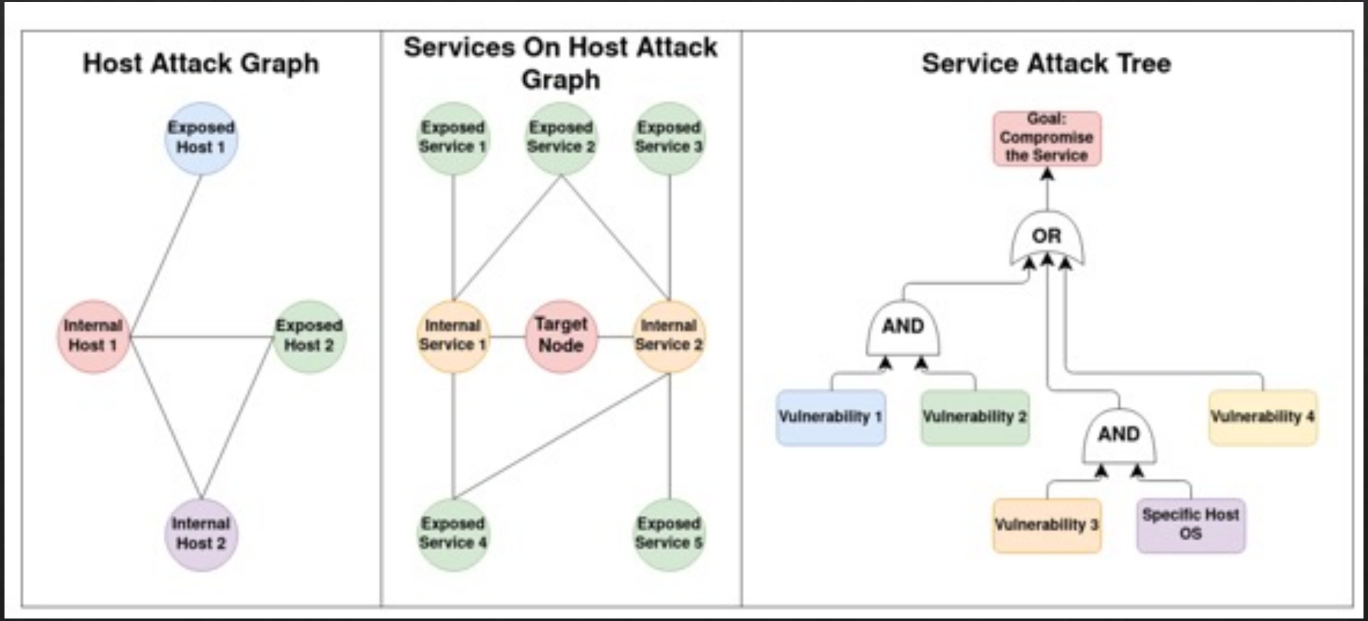
Attack Operation



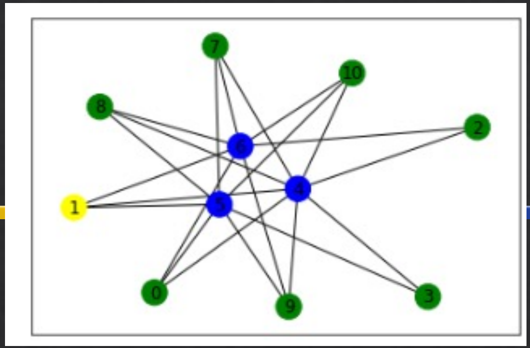
MTD operations



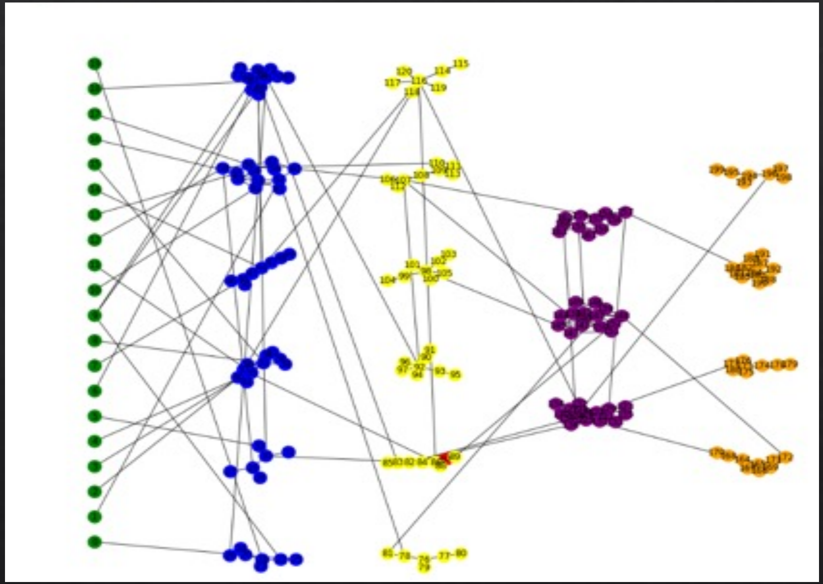
The Model – 3-Layer HARM



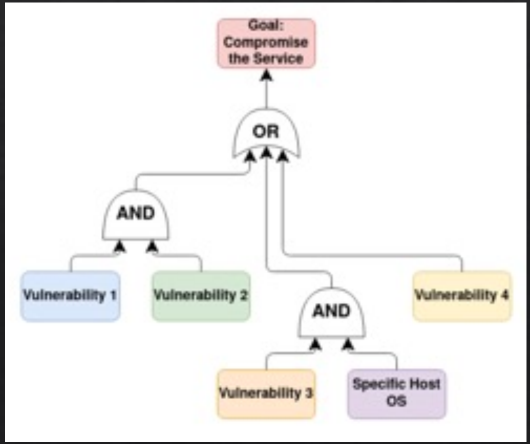
System Modeled



Service Attack Graph – Watts-Strogatz

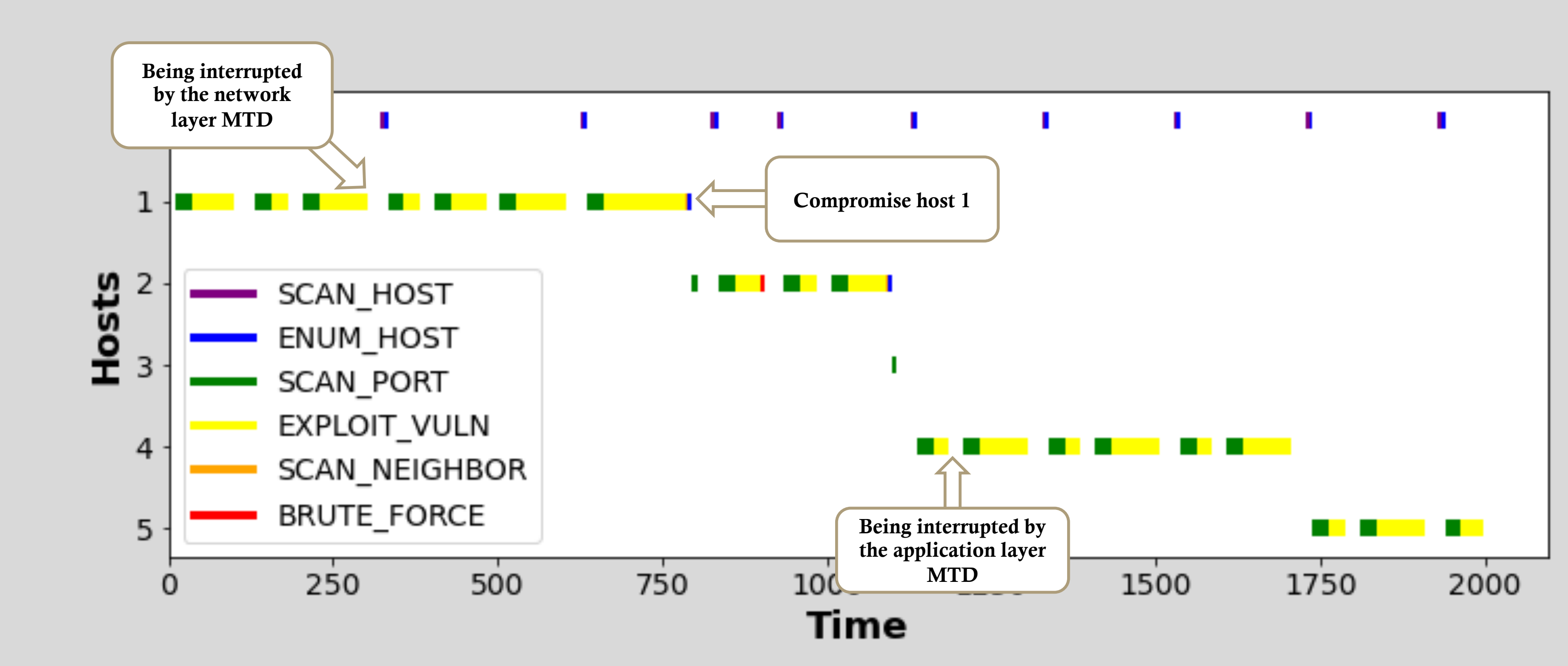


Host Attack Graph - Barabasi-Albert



Vuln Attack Tree – Conditional

Attack operation visualization



MTD Demo

Summary

- MTD is one of proactive security solutions that changes the attack surface continuously
- Various MTD techniques have been proposed, applicable in multiple system levels
- It is still difficult to determine how to optimally deploy MTD techniques in given systems and networks
- Using security modelling and analysis can be useful to understand MTD techniques and their effects in systems and networks
- Use of AI and ML can assist in configuring MTD deployment strategies

Additional reading

- Cho, Jin-Hee, Dilli P. Sharma, Hooman Alavizadeh, Seunghyun Yoon, Noam Ben-Asher, Terrence J. Moore, Dong Seong Kim, Hyuk Lim, and Frederica F. Nelson. "Toward proactive, adaptive defense: A survey on moving target defense." IEEE Communications Surveys & Tutorials 22, no. 1 (2020): 709-745.